Questions

Q1.

A light wave from a star has a frequency of 6.67×10^{14} Hz and a wavelength of 4.50×10^{-7} m.

The star is 4.00×10^{16} m away from Earth.

Calculate the time it takes light from the star to reach the Earth.

(3)

time to reach Earth = ... s

Q2.

* Explain the differences between longitudinal and transverse waves.
Your explanation should refer to ultraviolet, ultrasound and seismic waves.

(6)

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Q3.

Figure 7 shows a tuning fork.

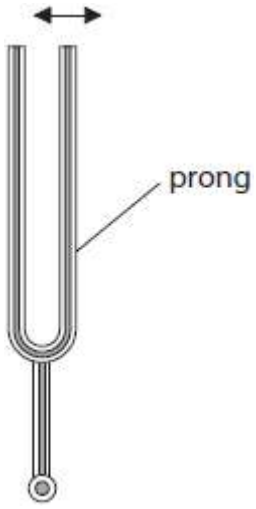

Figure 7

When the prongs of the tuning fork are struck, the prongs vibrate in the directions shown by the arrows on Figure 7.

Describe how the vibrating tuning fork causes a sound wave to travel through the air.

You may add to the diagram if it helps your answer.

(2)

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(Total for question = 2 marks)

Q4.

Ultrasound from a fishing boat is used to find fish.
(a) (i) Which of these is correct for ultrasound waves?
Put a cross (■) in the box next to your answer.

(1)

☒ **A** ultrasound waves have a frequency above 20 000 Hz

☒ **B** ultrasound waves have a frequency below 20 Hz

☒ **C** ultrasound waves have a wavelength above 20 000 Hz

☒ **D** ultrasound waves have a wavelength below 20 Hz

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The system that uses ultrasound to find fish is called

(1)

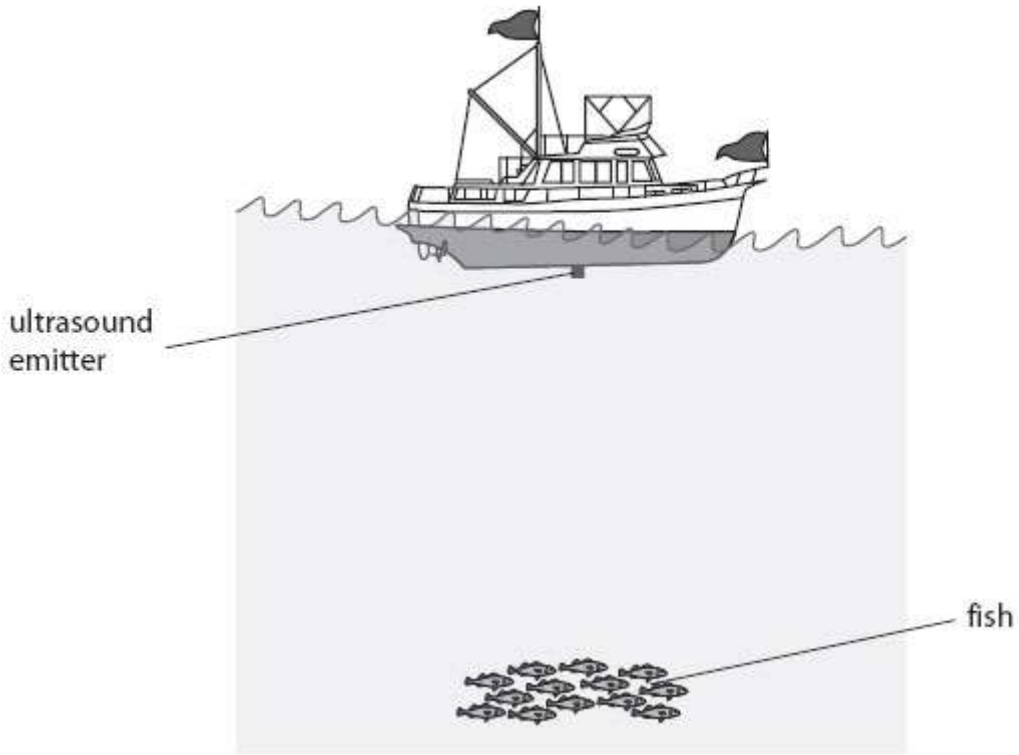
☒ **A** fibre optics

☒ **B** satellite transmission

☒ **C** sonar

☒ **D** thermal imaging

(iii) The diagram shows a fishing boat above some fish.



Describe how ultrasound waves are used to detect the fish.

You may add to the diagram to help with your answer.

(2)

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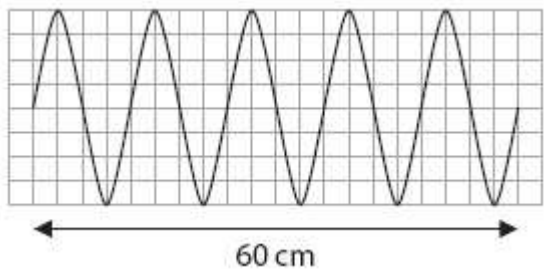
(b) Some students are investigating waves.
They produce waves by moving a piece of wood up and down in a tank of water.
The diagram shows the waves over a distance of 60 cm.



(i) State the number of wavelengths shown on the diagram.

(1)

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(ii) Calculate the wavelength of the waves.

(1)

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(c) The students produce a different wave.
This wave has a frequency of 1.7 Hz and a wavelength of 8.0 cm.

Calculate the speed of this wave.

(2)

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(Total for Question = 8 marks)

Q5.

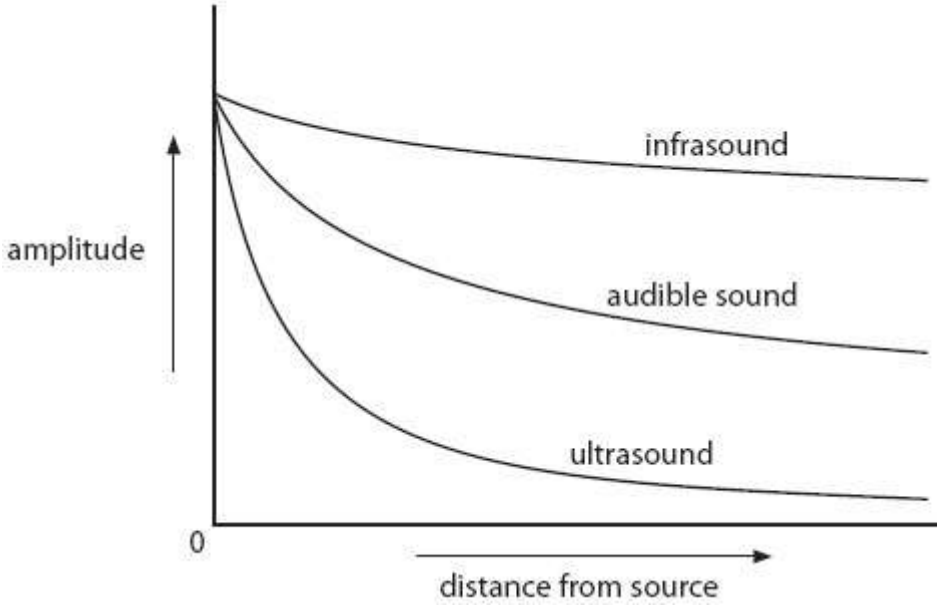
(a) Dolphins emit sounds at many different frequencies.

(i) Complete the sentence by putting a cross (■) in the box next to your answer.

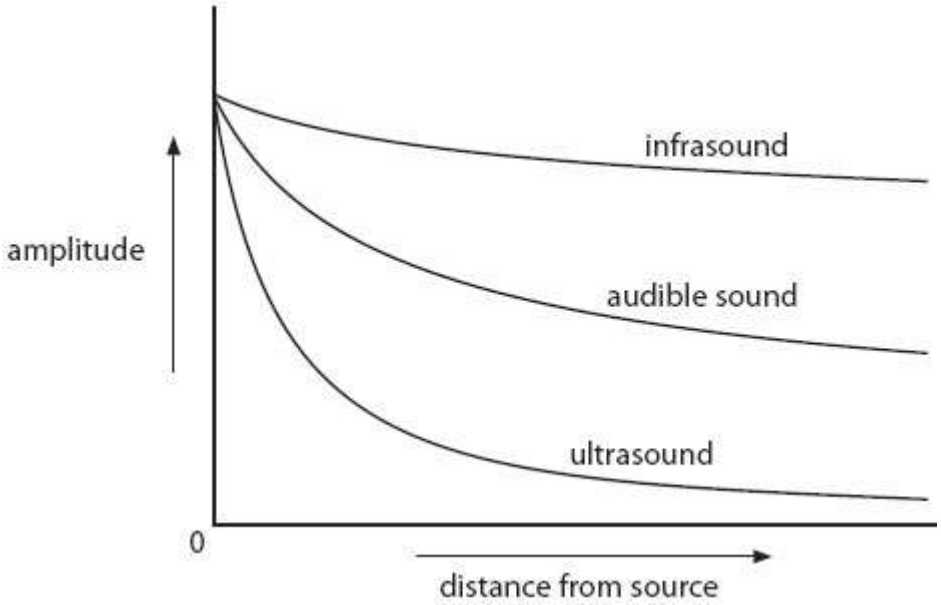
A dolphin can produce sounds with a frequency of more than 20 kHz.

These sounds are called

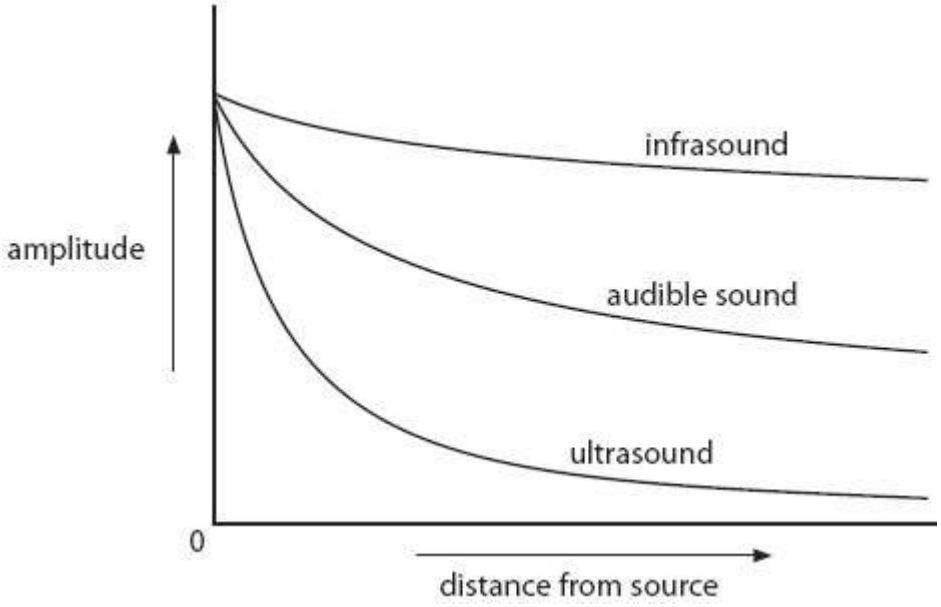
(1)



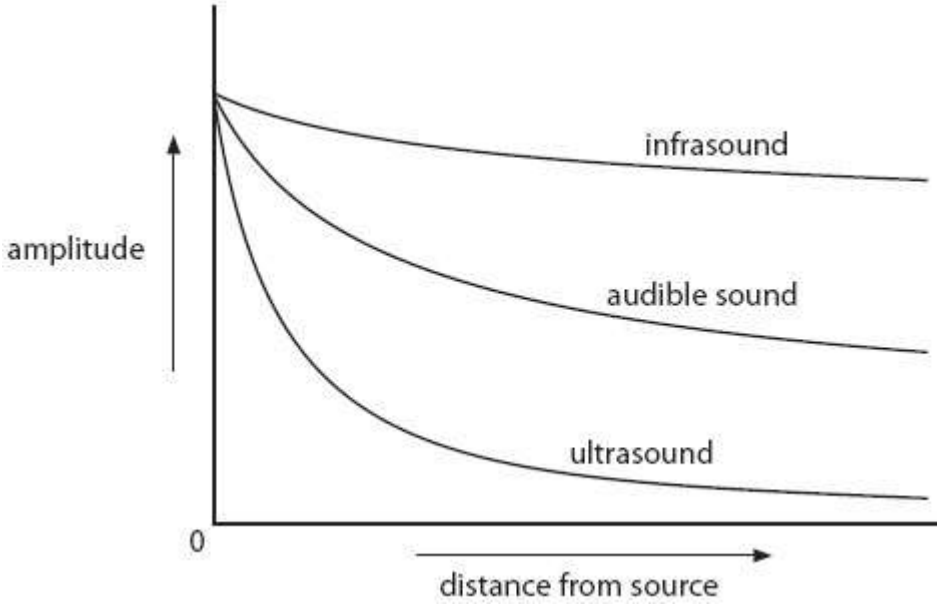
A infrasound waves



B microwaves



C supersonic waves



D ultrasound waves

(ii) A dolphin is near to a group of fish.

Describe how a dolphin uses high frequency sound to locate the position of the fish.

You may add to the diagram to help with your answer.

(3)

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(b) As a sound wave travels through water its amplitude gets smaller.
This graph shows how the amplitude of different types of sound waves decreases as they travel through water.

Whales use sound to communicate over long distances.
Explain which is the best type of sound wave for whales to use when communicating over long distances.

(2)

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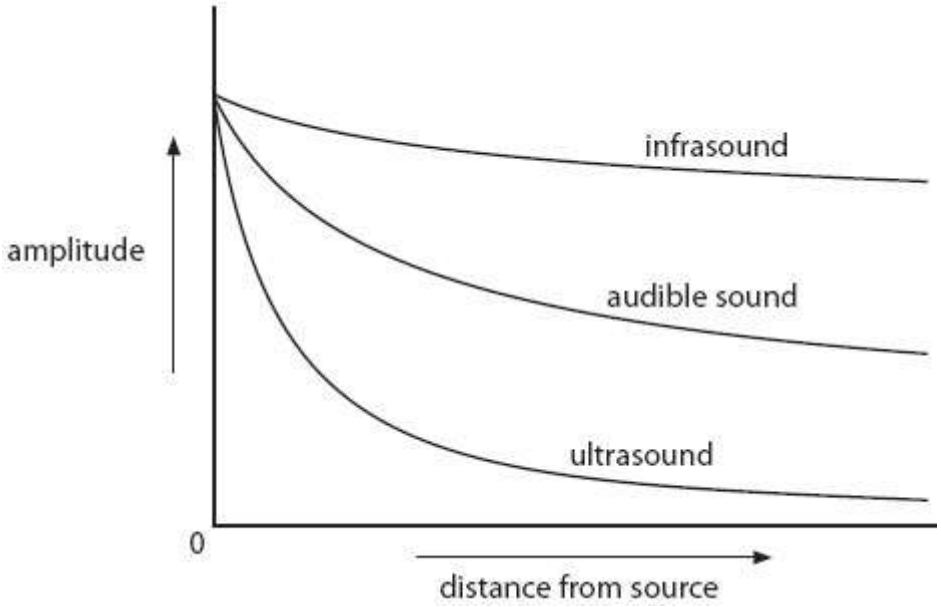
(c) Geologists use sound waves from a small explosion to search for oil underground.

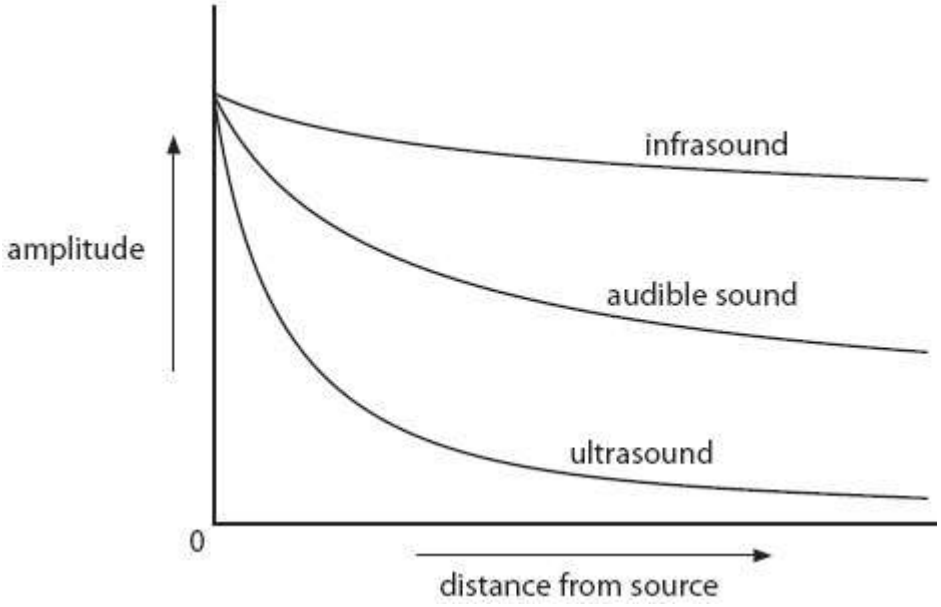
Complete the sentence by putting a cross (■) in the box next to your answer.

(i) These sound waves are called

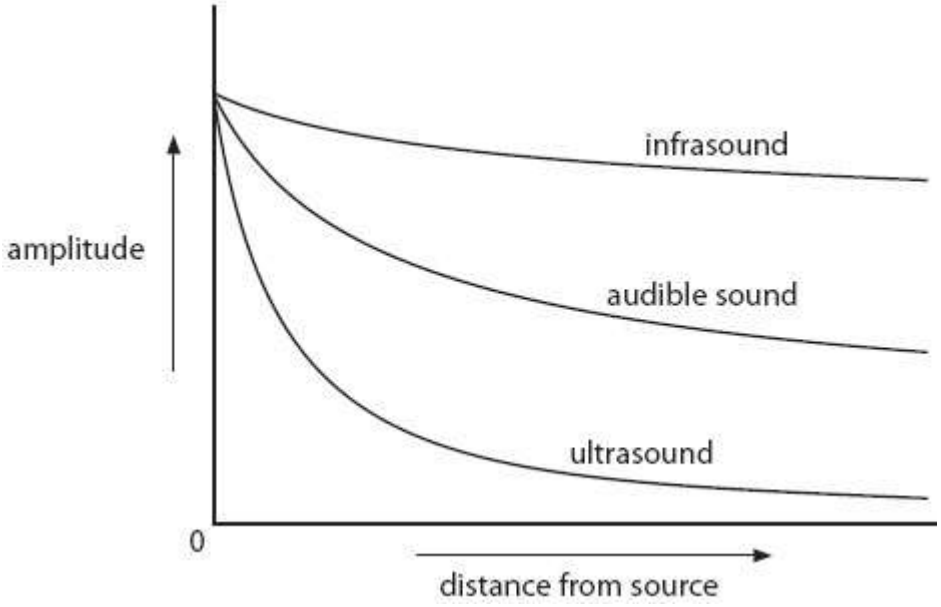
(1)



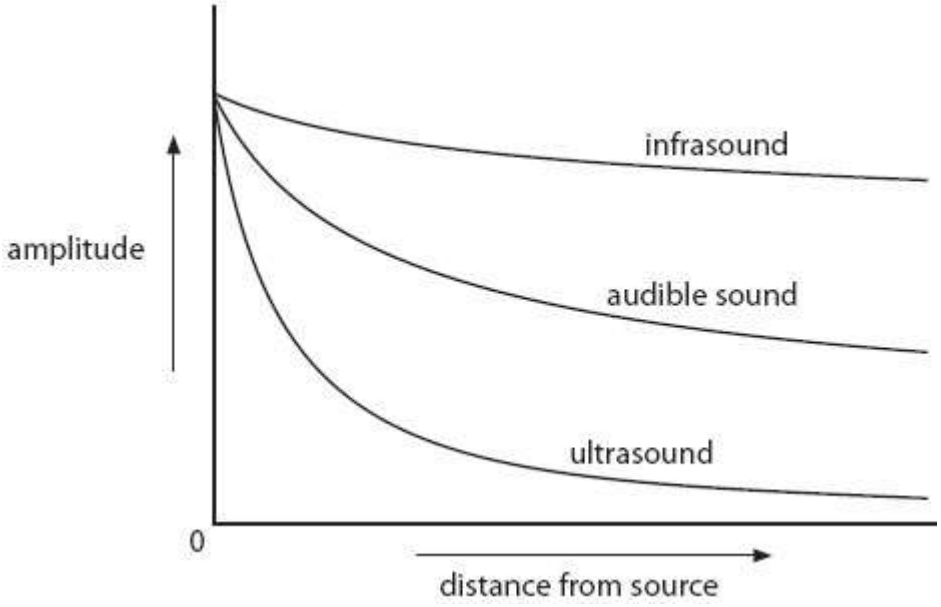
A cosmic waves



B seismic waves



C volcanic waves



D tectonic waves

(ii) A small explosion is triggered at the Earth's surface.
The waves reflect back from the top of the oil field.

Suggest why the waves are reflected from the oil field.

(1)

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(d) A wave has a frequency of 15 Hz.
Its wavelength is 125 m.

Calculate the speed of the wave.

State the unit.

(3)

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(Total for Question is 11 marks)

Q6.

Sonar is an example of a use of ultrasound.

State **one** other example of a use of ultrasound.

(1)

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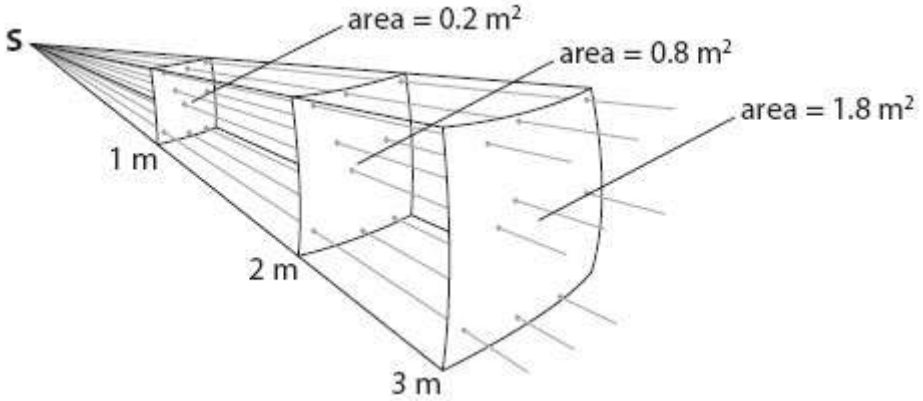
(Total for question = 1 mark)

Q7.

The diagram shows light from a point source, **S**, spreading out as it gets further from **S**.



(a) The intensity of light passing through the surface which is 1 m from **S** is 2.5 W/m^2.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The intensity of light, in W/m^2, passing through the surface which is 2 m from **S** is

(1)

☐ **A** 2.5 ÷ 2

☐ **B** 2.5 ÷ 4

☐ **C** 2.5 × 2

☐ **D** 2.5 × 4

(ii) Calculate the power of the light passing through the surface which is 1m from **S**.

(2)

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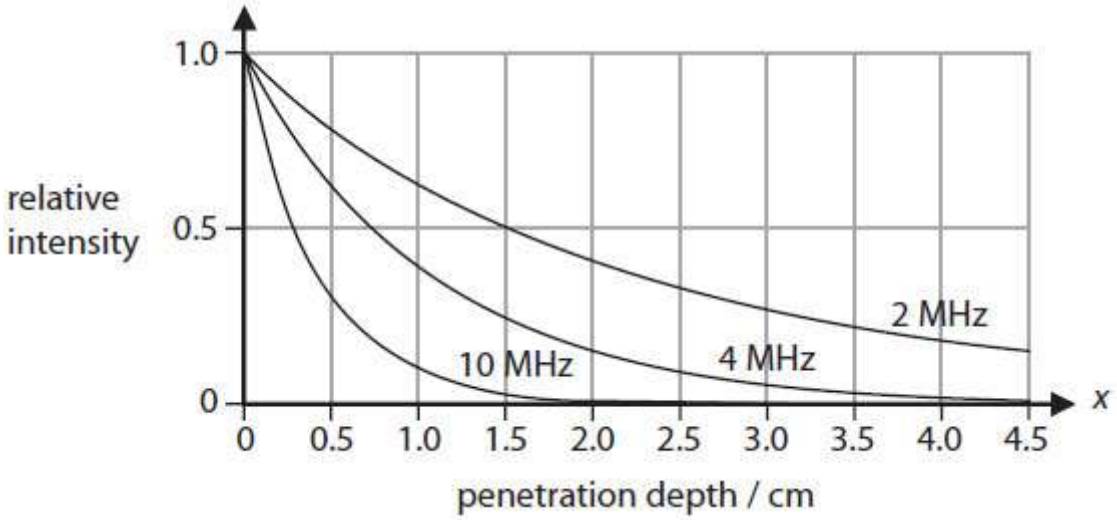
(b) The graph shows how the intensity of ultrasound waves of different frequencies decreases as they penetrate soft tissue.



(i) Estimate how far a 2 MHz wave has penetrated into the soft tissue when its intensity is 25% of its original value.

(1)

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(ii) Explain which of these frequencies of ultrasound can be used to scan organs deep inside the body.

(2)

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*(c) Medical physicists have developed endoscopes and many other devices to help doctors diagnose medical problems.

Compare the use of electromagnetic radiation in endoscopes and in one other diagnostic device.

(6)

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(Total for Question = 12 marks)

Q8.

Sonar is an example of a use of ultrasound.

State an example of a use of infrasound.

(1)

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(Total for question = 1 mark)

Q9.

Figure 2 is a diagram of a water wave.

A cork is floating on the water.

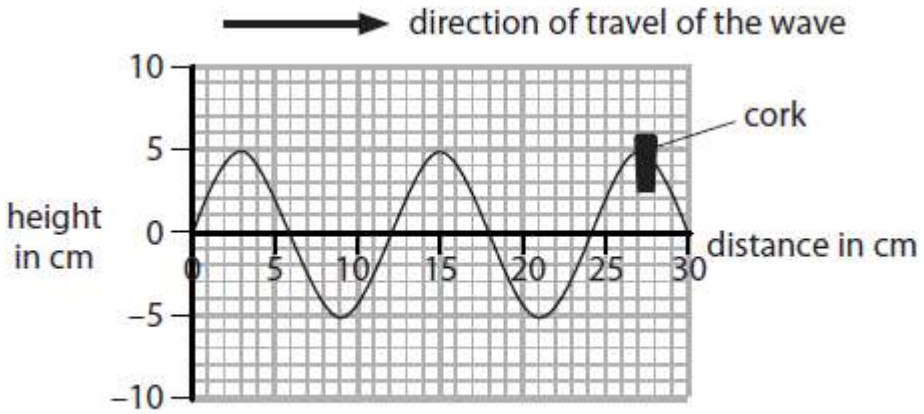

Figure 2

(i) Use the scale on the diagram to measure the wavelength of the wave.

(2)

wavelength = ... cm

(ii) Describe the motion of the cork.

You should include how the cork moves relative to the direction of travel of the wave.

(2)

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(Total for question = 4 marks)

Q10.

Sound travels slower in cold air than it does in warm air.

The equation relating the speed of sound in air to the density of the air is

$$\text{speed of sound} = \frac{K}{\sqrt{(\text{density})}} \qquad \text{where K is a constant.}$$

The table in Figure 10 gives some data about the speed of sound in air and the density of air.

	speed of sound in m/s	density of air in kg / m³
in cold air	331	1.29
in warm air		1.16

Figure 10

Use the equation and the data in the table in Figure 10 to calculate the speed of sound in warm air.

Give your answer to an appropriate number of significant figures.

(3)

speed of sound in warm air = ... m/s

(Total for question = 3 marks)

Q11.

Ultrasound from a fishing boat is used to find fish.

(i) Which of these is correct for ultrasound waves?

Put a cross (■) in the box next to your answer.

(1)

☐ **A** ultrasound waves have a frequency above 20 000 Hz

☐ **B** ultrasound waves have a frequency below 20 Hz

☐ **C** ultrasound waves have a wavelength above 20 000 Hz

☐ **D** ultrasound waves have a wavelength below 20 Hz

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.

The system that uses ultrasound to find fish is called

(1)

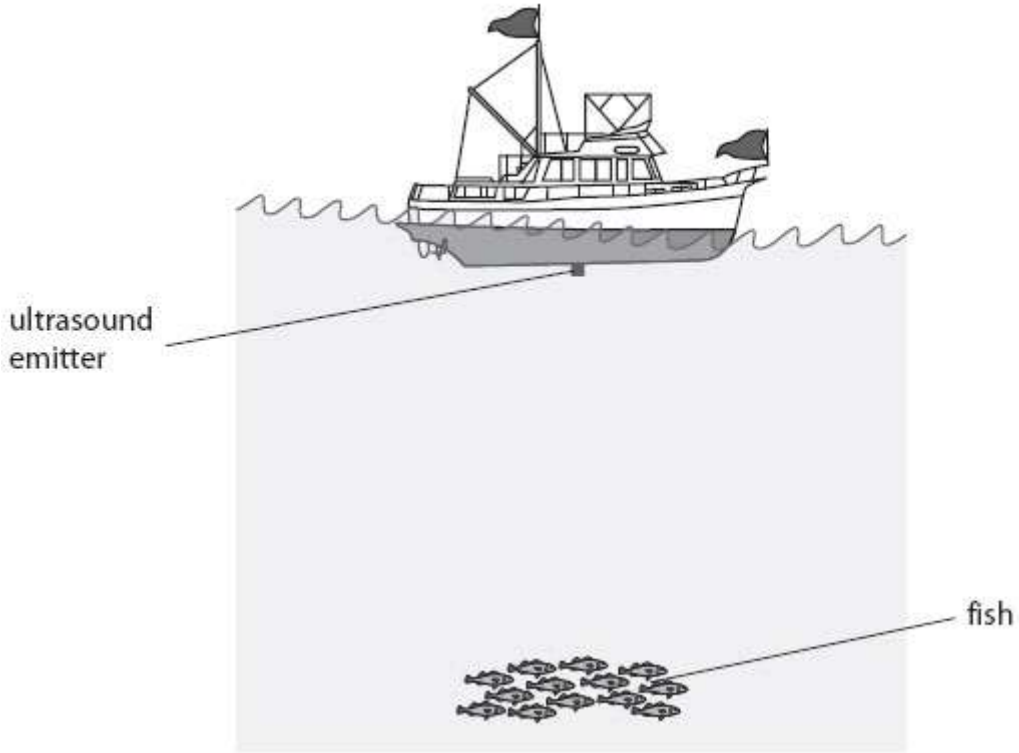
☐ **A** fibre optics

☐ **B** satellite transmission

☐ **C** sonar

☐ **D** thermal imaging

(iii) The diagram shows a fishing boat above some fish.



Describe how ultrasound waves are used to detect the fish.

You may add to the diagram to help with your answer.

(2)

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Q12.

This question is about ultrasound.

Ultrasound has many uses.

(i) One device called a pest repeller emits ultrasound.

The ultrasound keeps mice out of the garden.
Explain why the device affects mice but does not affect humans.

(2)

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(ii) A technician has a different ultrasound device.

This device can emit and detect short pulses of ultrasound.
The device can also measure the time, in ms, from emitting a pulse to detecting the same pulse.
Describe how the technician can use this device to determine the speed of ultrasound in air.

(3)

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(Total for question = 5 marks)

Mark Scheme

Q1.

Question Number	Answer	Acceptable answers	Mark
	Substitution (1) (Speed =) $6.67 \times 10^{14} \times 4.5 \times 10^{-7}$ Transposition AND substitution (1) (time =) $\frac{4 \times 10^{16}}{(6.67 \times 10^{14} \times 4.5 \times 10^{-7})}$ Evaluation (1) 1.33×10^{8} (s)	Award full marks for correct answer with no working 3×10^{8} (m/s) seen anywhere $\frac{4 \times 10^{16}}{3 \times 10^{8}}$ ECF candidate's speed maximum 2 marks Allow answers which round to 130 000 000 IGNORE Power of Ten error until evaluation	**(3)**

Q2.

Question Number		Indicative Content	Mark
QWC	*	An explanation including some of the following points • Longitudinal {vibrations/oscillations} are {along/parallel to/in the same direction as} the direction of {travel/energy transfer} • Transverse {vibrations/oscillations} are {across/perpendicular to/90° to/right angles to} the direction of {travel/energy transfer} • Ultraviolet waves are transverse • Ultrasound waves are longitudinal (ignore sound – not on list) • Some seismic waves are longitudinal and some are transverse • P waves are longitudinal • S waves are transverse • Longitudinal waves need a material for the vibrations whereas electromagnetic waves can pass through a vacuum **IGNORE** irrelevant material	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited explanation of: EITHER the {vibration/movement} direction and direction of {travel/movement} for transverse or longitudinal wave OR correctly identifying the wave type for at least one example from the list, e.g. o Longitudinal waves move in the same direction as the wave moves o Ultraviolet waves are transverse • the answer communicates ideas using simple language and uses limited scientific terminology
2	3 - 4	• a simple explanation linking: EITHER directions of {vibration/oscillation} and wave travel for both types of wave OR direction of {vibration/oscillation} and wave travel of one type of wave with at least one example of a wave from the list OR the direction of 'movement' and direction of {travel/movement} for transverse AND longitudinal waves AND correctly identifying the wave type for at least one example from the list e.g. o In longitudinal waves the vibrations are in the same direction as the wave travels. In transverse waves the vibrations are at right angles to the direction the wave travels. o In longitudinal waves the vibrations are in the same direction as the wave travels. Ultraviolet waves are transverse. o Longitudinal waves move in the same direction as the wave moves. Transverse waves move at right angles to the direction the wave moves. Ultrasound waves are longitudinal. • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	• a detailed explanation clearly differentiating between the directions of {vibration/oscillation} for longitudinal AND transverse waves AND at least one example of each type of wave from the list, e.g. o In longitudinal waves the vibrations are in the same direction as the wave travels. In transverse waves the vibrations are at right angles to the direction the wave travels. Ultrasound waves are longitudinal and ultraviolet waves are transverse. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q3.

Question Number	Answer	Additional guidance	Mark
	a description to include: (the prong makes the) air vibrate/oscillate (1) in the same direction as /parallel to the wave travels (1)	causes compressions and rarefactions in air transfers ke to air longitudinal credit can be given for a labelled diagram	**(2)** AO 1 1

Q4.

	Answer	Acceptable answers	Mark
(a)(i)	☒ **A** ultrasound waves have a frequency above 20 000 Hz		**(1)**
(a)(ii)	☒ **C** sonar		**(1)**
(a)(iii)	a description including any **two** of the following: • (ultrasound waves / pulses) go down (through the water) (1) • (ultrasound waves are) reflected **off fish** (1) • (reflected ultrasound waves) are received by boat (1) • time delay (shows how deep fish are) (1)	on diagram, wave or ray indicated as downwards idea of wave moving towards or hitting fish on diagram, waves or rays reflected **off fish** idea of wave bouncing **off fish** signal is timed **ignore** fish emitting ultrasound	**(2)**
(b)(i)	(number of waves =) 5 (1)		**(1)**

(b)(ii)	**Either** 60 ÷ 5 (1) **or** 60 ÷ (their answer to 2(b)(i)) (1)	12 (cm) **or** ecf from number of waves	**(1)**
(c)	Substitution 1.7 × 8 (1) Evaluation 14 (cm/s) (1)	13.6 (cm/s) give full marks for correct answer, no working Power of 10 error max. 1 mark.	**(2)**

Q5.

	Answer	**Acceptable answers**	**Mark**
(ai)	D ultrasound waves (1)		**(1)**
(aii)	Any 3 from • emits (high frequency/ultra) (sound)(1) • (sound is) reflected (off fish) (1) • (reflection) detected by Dolphin (1) • dolphin (estimates) time between (sending and receiving) sounds (1) • dolphin is able to change time into (estimate of) distance (1)	Makes/sends out/produces (ultra sound/signal/wave) Uses 'high frequency sound' is insufficient (sound) bounces off (fish) or echoes towards dolphin 1st three marks can be scored on the diagram. ie unless stated otherwise, assume any waves/rays starting at dolphin are ultrasound. Rays do not need to be straight	**(3)**
(b)	An explanation including: • Infrasound (1) Plus one from: • Decrease/change in amplitude is least (1) • can be detected/'heard' further away (1)	Marks are independent Stays the biggest/stays high. Has a bigger amplitude would travel the furthest/further	**(2)**
(ci)	B seismic waves (1)		**(1)**
(cii)	(there is a) difference/change in density (1)	more/less/too dense (reach a) boundary (between different materials) Ignore 'the waves cannot travel through liquids/oil'	**(1)**
(d)	Substitution into correct	Power of 10 error max 1 mark for	**(3)**

	equation(1) v = 15 × 125 Evaluation (1) 1875 Unit (1) m/s	numerical answer 2 marks for correct numerical answer even with no working shown ms^{-1} **not** mps 1.875 km/s or 6750 km/h gain 3 marks If numerical answer incorrect, accept any correctly-written unit of speed: eg **k**m/s or km/hr or miles per hour / mph	

(Total for Question = 11 marks)

Q6.

Question Number	Answer	Additional Guidance	Mark
	foetal scanning (1)	ignore sonar ACCEPT echo location accept other examples e.g. dog whistle cat scarer bat detector kidney stones cleaning jewellery baby scanner pregnancy scanner faults in structures	**(1)** AO 1 1

Q7.

	Answer	Acceptable answers	Mark
(a)(i)	■ **B** 2.5 ÷ 4		**(1)**
(a)(ii)	either P = 2.5 × 0.2 or 2.5 = P / 0.2 (1) 0.5	give full marks for correct answer, no working	**(2)**

	(W) (1)		
(b)(i)	3.0 +/- 0.5 (cm)		**(1)**
(b)(ii)	an explanation linking • 2 MHz (1)and any **one** from: • has a higher intensity inside tissue (1) • less energy absorbed (1) • less attenuation (1) • penetrates furthest /deepest (1)	this frequency alone RA loses intensity more gradually highest penetration accept "2MHz and 4MHz" with correct reason for 1 mark	**(2)**

		Indicative Content	**Mark**
QWC	***(c)**	A comparison of endoscopes with any one of the following devices: **Diagnostic devices** • CAT scanners • Fluoroscopes • Thermal imagers / IR thermometers • Pulse oximeters • PET scanners • X-ray machines • Gamma cameras**Link to electromagnetic radiation** • Endoscopes use TIR of light in optical fibres • CAT scanners X-rays and computer to generate 3D images • Fluoroscopes use X- rays and a video camera • Thermal imagers use infrared emitted by a body • IR / red LEDs used to measure oxygen levels • PET scanners detect radiation emitted by electronpositron annihilation • Gamma cameras detect gamma rays	**(6)**

		from radioactive sources**Other factors for comparison** • Safety • Ease of use • Frequency / wave length • Intensity • Penetration • Ionising / non-ionising	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited comparison between an endoscope and one device e.g. endoscopes use light and CAT scanners detect broken bones • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple comparison between an endoscope and one device, linking them to the electromagnetic radiation used for both and a detail of use for one of them e.g. endoscopes use visible light to examine internal organs and CAT scans use X-rays • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed comparison between an endoscope and one device, linking them to the electromagnetic radiation used for both and a detail of use for both of them e.g. endoscopes use visible light which is passed down optical fibres by TIR to examine internal organs. Fluoroscopes use X-rays and a video camera to show positioning of stents in arteries. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q8.

Question Number	Answer	Additional Guidance	Mark
	studying the Earth's structure (1)	accept other examples e.g. detect meteor(ites) seismic activity named animals communicating e.g. elephants giraffes whales	**(1)** AO 1 1

Q9.

Question Number	Answer	Additional guidance	Mark
(i)	evidence of use of scale on horizontal distance axis only (1) 12 (cm) (1)	may be seen on the diagram range 11.5 to 12.5 (cm) award full marks for the correct answer without working 6 (cm) or 30(cm) scores 1 mark (evidence of use)	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	a description to include: moves up and down (1) at right angles / normal / perpendicular to (direction of) wave / travel (1)	independent marking points vertical (oscillations) not in the (direction of) wave / travel accept 'transverse wave' for 2nd MP	**(2)**

Q10.

Question Number	Answer	Additional guidance	Mark
	using cold row: evaluate (K=)376 (1) using warm row: substitute K and ρ $\frac{376}{\sqrt{1.16}}$ OR 349.10.... (1) 349 (m/s) to 3 sig figs (1)	 $\frac{\text{other K from earlier calculation}}{\sqrt{1.16}}$ any answer to 3 sig figs 349.10... scores MP1 and MP2 award full marks for the correct answer without working	**(3)**

Q11.

	Answer	Acceptable answers	Mark
(i)	☒ **A** ultrasound waves have a frequency above 20 000 Hz		**(1)**
(ii)	☒ **C** sonar		**(1)**
(iii)	a description including any **two** of the following: • (ultrasound waves / pulses) go down (through the water) (1) • (ultrasound waves are) reflected **off fish** (1) • (reflected ultrasound waves) are received by boat (1) • time delay	on diagram, wave or ray indicated as downwards idea of wave moving towards or hitting fish on diagram, waves or rays reflected **off fish** idea of wave bouncing **off fish** signal is timed **ignore** fish emitting ultrasound	**(2)**

	(shows how deep fish are) (1)		

Q12.

Question number	Answer	Additional guidance	Mark
i	An explanation linking: frequency (1) in mouse hearing range but not in human hearing range (1)	 accept wavelength relevant frequency value with unit eg>20 kHz	**(2) AO2**

Question number	Answer	Additional guidance	Mark
ii	A description to include: send pulse to a wall/reflecting surface OR detect the echo (1) measure distance (to wall and time to echo) (1) use speed = 2 x $\frac{\text{distance}}{\text{time}}$ (1)	 accept measure time to receive echo OR reflection back OR record time to echo/return	**(3) AO3**